Combinatorics 1

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Goal

To understand:

- Binomial Coefficients
- Optimal Ways for Computing C(n, r)
- Arrangements

Binomial Coefficients

- Number of ways to choose K items from N items.
- https://en.wikipedia.org/wiki/Binomial_coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

$$\binom{n}{k} = \frac{n \times (n-1) \times \cdots \times (n-k+1)}{k \times (k-1) \times \cdots \times 1}$$

${}^{n}C_{K}$

$\binom{n}{K}$

${}^{5}C_{2}$

no. of combinations if you choose K items out of n items

Examples

- You are organizing a birthday party and have 15 gift options. You want to select 10 gifts to give away as party favors. Find the number of combinations. → ${}^{15}C_{10}$
- A school is organizing a football tournament. There are 20 students who volunteered to play, but the team can only have 11 players. Find the number of combinations. → ${}^{20}C_{11}$

$$\overbrace{O \quad O \quad - - - \quad O}^{15 \text{ gifts}}$$

choose 10

1st gift → 15

2nd gift → 14

3rd gift → 13

⋮

10th gift → 6

$$\frac{15 * 14 * 13 * \ldots * 6}{10!}$$

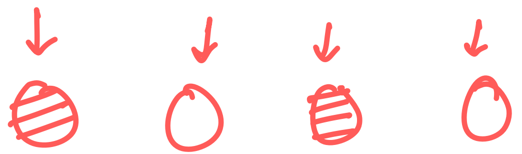
$n = 4$ $\quad$ $K \to 2$

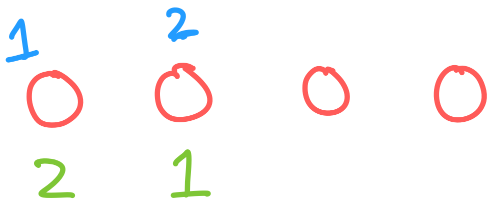
12 $\quad$ 13 $\quad$ 14 $\quad$ 23 $\quad$ 24 $\quad$ 34

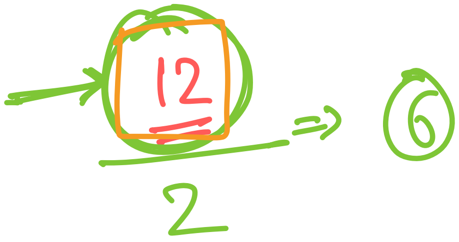
21 $\quad$ 31 $\quad$ 41 $\quad$ 32 $\quad$ 42 $\quad$ 43

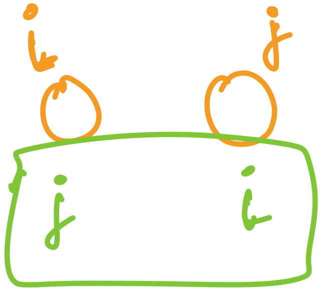
1st gift → 4

2nd gift → 3









every unique combination must be counted i j

○ ○

j i

$$\overbrace{\bigcirc \quad \bigcirc \quad \cdots \quad \bigcirc}^{r}$$

r!

$$\left\{\begin{array}{ccc} a & b & c \\ \bigcirc & \bigcirc & \bigcirc \\ a & c & b \\ b & a & c \\ b & c & a \\ c & a & b \\ c & b & a \end{array}\right.$$

$n \qquad r$

$1st \to n$

$2nd \to n-1$

$3rd \to n-2$

$\vdots$

$r^{th} \to n-r+1$

$${}^{n}P_{r} = \underline{n * (n-1) * (n-2) * \ldots * (n-r+1)}$$

$$n!/(n-r)!$$

$${}^{n}C_{r} = \frac{n * (n-1) * \ldots * (n-r+1)}{r!}$$

$$\boxed{{}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}}$$

How to Compute C(n, r)

$${}^{n}C_{r} = \frac{n!}{r! * (n-r)!} \rightarrow \text{too huge}$$

n = 100
r = 50

- **C(n, r) = fact(n) / (fact(r) * fact(n - r))**.
- Since the value might be too huge, generally we are asked to calculate **C(n, r) % P**. → prime number → 10^9+7
- We know that **A^P % P = A % P** (Fermat's Theorem)
  - It can be rewritten as **(1 / A) % P => A^{P-2} % P**.
- Let's define **inv(A) = A^{P-2} % P.**
- Therefore **C(n, r) % P = fact(n) * inv(fact(r)) * inv(fact(n - r)) % P**.

$$\frac{A^P \% P}{A^2} = \frac{A \% P}{A^2}$$

$$\boxed{(1/A) \% P = A^{P-2} \% P}$$

$$\boxed{(A/B) \% P \Rightarrow A * B^{P-2} \% P}$$

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$$^{n}C_{r} \,\%\, p = \underline{\underline{n!}} * \underline{\underline{inv}}(\underline{\underline{r!}}) * \underline{\underline{inv}}(\underline{\underline{(n-r)!}}) \,\%\, p$$

$$\underline{inv(a) = a^{p-2} \,\%\, p}$$

Requirements to Calculate C(n, r)

- Mod Inverse Calculation
- Precomputation of Factorials
- Precomputation of Inverse Factorials

binary expo

$inv(a) = a^{p-2} \% p$

$O(\log P)$

$max\_n = 10^5$

fact(n) for all $1 \le n \le 10^5$

$fact[n] = (1*2*3*..*n) \% p$

$inv\_fact[n] = \left(\frac{1}{fact[n]}\right) \% p = inv(fact[n])$

$C(n,r) = fact[n] * inv\_fact[r] * inv\_fact[n-r] \% p$

How to Compute C(n, r)

- Calculate **fact[n] = (factorial of n) % P** for all **0 <= n <= max_n**:
  - fact[0] = 1
  - fact[i] = fact[i - 1] * i % P
  - TC -> O(max_n)
- Calculate **inv_fact[n] = (1 / fact[n]) % P** for all **0 <= n <= max_n**:
  - inv_fact[n] = inv(fact[n]) for all 0 <= n <= max_n
  - TC -> (max_n * log(~~max_n~~))

$n! = (n-1)! * n$

TC → $\boxed{O(max\_n * \log P)}$

inv_fact[0]
[i]
: [max_n]

How to Compute C(n, r)

- We can also compute the inv_fact[i] for all i from 0 to max_n in O(max_n):
    - inv_fact[max_n] = inv(fact[max_n]) → O(log P)
    - inv_fact[i] = inv_fact[i + 1] * (i + 1) % P

$O(\log P)$

$O(1)$

$O(\text{max\_n} + \log(P))$

$O(\text{max\_n} + \log(P))$

$$\frac{1}{n!} = \frac{1}{(n+1)!} * (n+1) \Rightarrow \text{inv\_fact}[n] = \text{inv\_fact}[n+1] * (n+1)$$

$\underline{inv(a)} = \boxed{a^{p-2} \% p}$

$O(\log(p))$

$a^b \% M \Rightarrow$ binary expo in $O(\log(b))$

$O(N)$

$O(max\_n)$

How to Compute C(n, r)

Time Complexities:

- Precomputation:
  - O(max_n) for fact[n] ✓
  - O(log(~~max_n~~ P)) for inv_fact[max_n] → O(log P) due to inv()
  - O(max_n) for inv_fact[n] ✓
  - Overall -> **O(max_n)** + O(log P)
- Per query
  - **O(1)** ✓

$^nC_r \% P = fact[n] * inv\_fact[r] \% P * inv\_fact[n-r] \% P$

Code for C(n, r)

$(1LL * n) \Rightarrow n^{(\text{long long})}$

```
int fact[N + 1], inv_fact[N + 1];

void calculate() {  // O(N)
  fact[0] = 1;
  for (int i = 1; i <= N; i++) fact[i] = 1LL * fact[i - 1] * i % P;
  inv_fact[N] = inv(fact[N]);
  for (int i = N - 1; i >= 0; i--)
    inv_fact[i] = 1LL * inv_fact[i + 1] * (i + 1) % P;
}

int C(int n, int r) {  // O(1)
  if (r < 0 || r > n) return 0;
  return 1LL * fact[n] * inv_fact[r] % P * inv_fact[n - r] % P;
}
```

$N \to$ max-n

$(fact[n])^{P-2} \% P$

$\left(1/fact[n]\right) \% P$

Another way to compute C(n, r)

$$\binom{n}{k} = \frac{n^{\underline{k}}}{k!} = \frac{\overbrace{n(n-1)(n-2)\cdots(n-(k-1))}^{K \text{ integers}}}{\underbrace{k(k-1)(k-2)\cdots 1}_{K \text{ integers}}} = \prod_{i=1}^{k}\frac{n+1-i}{i}$$

2 different use cases:

- Calculate **C(n, r)** many times with **O(n)** precomputation and **O(1)** calculation of each query.
- Calculate **C(n, r)** just once in **O(r)** time.
  Example -> Let's say max_n <= 10^9 and max_r <= 20.

$\binom{n}{C_r}$

$n \rightarrow$ too huge

$r \rightarrow$ small

① $\boxed{{}^nC_r \% P = n * (n-1) * \ldots * (n-r+1) * inv(r!) \% P}$ ✓

② $${}^nC_r = \left[\left(\frac{n}{1} * (n-1)\right)/2 * (n-2)\right]/3 * \quad ✓$$

$$\frac{n * (n-1) * (n-2) * (n-3) * \cdots}{1 * 2 * 3 * 4 *}$$

```
C(n,r){
  res = 1;
  for (i = 1; i ≤ r; i++) {
      res *= n-i+1;
      res /= i;
  }
}
```

Important Binomial Results

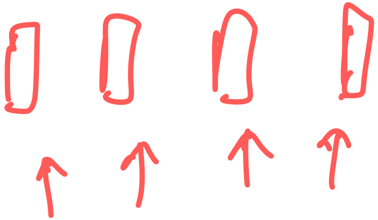
$$\binom{n}{0} = \binom{n}{n} = 1$$

n items

$\boxed{{}^{n}C_{0} = 1}$

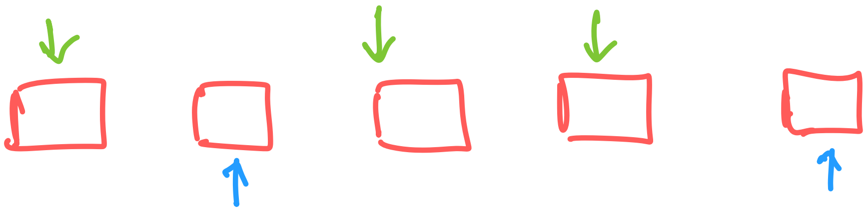
choose 0

$\boxed{{}^{n}C_{n} \Rightarrow 1}$



Important Binomial Results

$$\binom{n}{k} = \binom{n}{n-k}$$



→ number of ways to choose 3 of them and color them green

→ choosing 2 of the gifts and coloring all the gifts other than the chosen 2

$$\boxed{{}^nC_r} = \frac{n!}{r! * (n-r)!} = \frac{n!}{(n-\underbrace{(n-r)}_{s})! * \underbrace{(n-r)}_{s}!}$$

$r \Rightarrow n-(n-r)$

$s = (n-r)$

$$= \frac{n!}{(n-s)! * s!} = {}^nC_s$$

$$= \boxed{{}^nC_{n-r}}$$

$$\boxed{{}^nC_r = {}^nC_{n-r}}$$

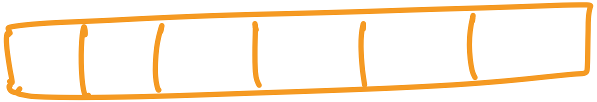
Made with Goodnotes

Important Binomial Results

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\boxed{{}^nC_0} + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

size n

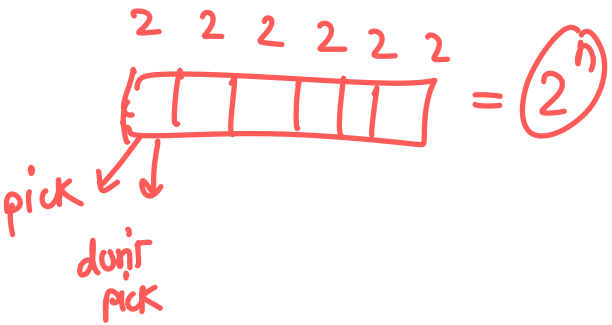


${}^nC_0 \Rightarrow$ no. of subsets of size 0

${}^nC_1 \Rightarrow$ no. of subsets of size 1

${}^{n}C_{i} \Rightarrow$ no. of subsets of size i in the array

$$\underbrace{{}^{n}C_{0} + {}^{n}C_{1} + \dots + {}^{n}C_{n}}_{\text{total no. of subsets in array}} \Rightarrow \boxed{2^{n} \text{ subsets}}$$



Binomial Expansion

$$(a+b)^n = {}^nC_0\, a^0 b^n + {}^nC_1\, a^1 b^{n-1} + {}^nC_2\, a^2 b^{n-2} + \dots + {}^nC_n\, a^n b^0$$

$a=1 \quad b=1$

$$(1+1)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$$

2^n

Important Binomial Results

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

^nC_k

n size

choose K

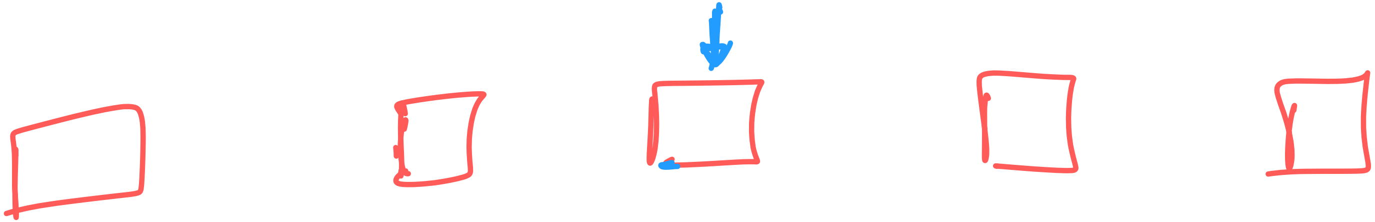
dont $\rightarrow$ $^{n-1}C_K$

pick $\rightarrow$ $^{n-1}C_{K-1}$

$$\boxed{^nC_K = {}^{n-1}C_{K-1} + {}^{n-1}C_K}$$

Important Binomial Results

- K items always included -> **C(n - k, r - k)**



fixed K items always included

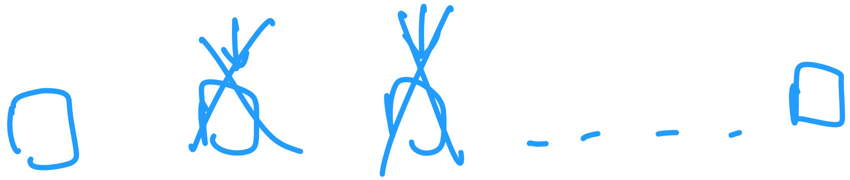
n items choose r

$(n-K) \rightarrow$ choose $(r-K)$

$^{n-K}C_{r-K}$

Important Binomial Results

- K items are never included -> **C(n - k, r)**



$^{n-k}C_r$

Important Binomial Results

- Value of **C(n, r)** is greatest when **r = n/2**
  - Example:
    - C(6, 0) = 1
    - C(6, 1) = 6
    - C(6, 2) = 15
    - C(6, 3) = 20 // C(6, r) is largest when r = 6 / 2
    - C(6, 4) = 15
    - C(6, 5) = 6
    - C(6, 6) = 1

$n \to$ even

${}^nC_{(n/2)}$

$n \to$ odd

${}^nC_{\lfloor n/2 \rfloor}$ and ${}^nC_{\lceil n/2 \rceil}$

$n=6$

${}^6C_3$

$n=5$

${}^nC_0 = 1$ ${}^nC_1 = 5$ ${}^nC_2 = 10$ ${}^nC_3 = 10$ ${}^nC_4 = 5$ ${}^nC_5 = 1$

Arrangement

- Arranging Distinct Elements -> **n!**
  - Find the number of permutations of string “abcd”.

$n \quad (n-1) \quad (n-2) \quad (n-3) \quad \_\_ \quad \_\_ \quad 1$

$n!$

Arrangement

$$\frac{(a_1 + a_2 + a_3)!}{a_1! * a_2! * a_3!}$$

- Arranging Similar Elements -> $(a_1 + a_2 + \ldots + a_n)! / (a_1! * a_2! * \ldots * a_n!)$
  - Find the number of permutations of string "aabccd".

aabccd → $a_1\ a_2\ b\ c_1\ c_2\ d$ → 6!

aabccd → one unique permutation

a → frequency 2

→ $a_1\ a_2\ b\ c_1\ c_2\ d$

→ $a_2\ a_1\ b\ c_1\ c_2\ d$

same

$a \rightarrow 2$

$2!$

aabccd

$a_1 a_2 b c_1 c d$

$a_2 a_1 b c c d$

$a_1 a_2 b c_1 c_2 d$

$a_1 a_2 b c_2 c_1 d$

aaab

$a \rightarrow 3$

repeat

$3!$

aaab

$a_1 a_2 a_3 b$

$a_1 a_3 a_2 b$

$a_2 a_1 a_3 b$

'a' $\rightarrow$ cnt_a

'b' $\rightarrow$ cnt_b

'c' $\rightarrow$ cnt_c

$\vdots$

$n = cnt_a + cnt_b + \ldots + cnt_z$

$$\boxed{\frac{n!}{(cnt_a)!\,(cnt_b)!\,(cnt_c)!\ldots(cnt_z)!}}$$

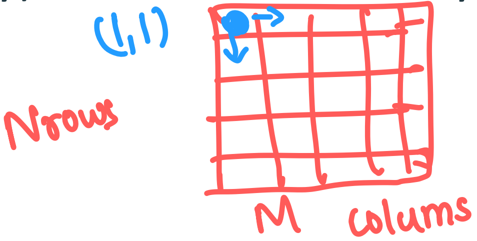
$\underline{aa}b\underline{cc}d \rightarrow \dfrac{6!}{2!\,2!}$

Problem: Unique Paths

Given a 2-d matrix with **N** rows and **M** columns. ($1 <= \mathbf{N, M} <= 10^5$)

Suppose, you're at cell (x, y), then in one move you can either move to

- (x + 1, y) down
- (x, y + 1) right



~~dp[x][y] → no. of ways to reach (x,y)~~

Starting from cell **(1, 1)**, find the number of ways of reaching the cell **(N, M)**.

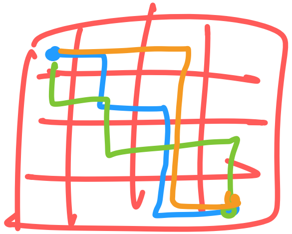
Since the answer might be too huge, answer it modulo $10^9 + 7$.

$\cancel{dp[x][y] = dp[x-1][y] + dp[x][y-1]}$

~~MLE~~

Problem Link - https://leetcode.com/problems/unique-paths

(1, 1) $\rightarrow$ (N, M)



D R

$\underline{\underline{(N+M-2)}}$

DRDRDDDR $\rightarrow$ different

DRDRDDDR $\rightarrow$ (N-1) times down

DRDDRDDR $\rightarrow$ (M-1) times right

number of arrangements of (N-1) D and (M-1) R

$$\frac{((N-1)+(M-1))!}{(N-1)!\,(M-1)!} \Rightarrow \frac{(N+M-2)!}{(N-1)!\,(M-1)!}$$

Made with Goodnotes

D R D R R D R D R D R $(N-1)$

$(N+M-2)$ steps

$${}^{N+M-2}C_{N-1}$$

$${}^{N+M-2}C_{M-1}$$

$$\frac{(N+M-2)!}{(N-1)!\,(M-1)!}$$

Problem: Creating String II

Given a string **S** of length **N** ($1 <= N <= 10^5$).

Find the number of unique strings that can be formed by rearranging the characters of the string **S**.

rearranging similar elements

Problem Link - https://cses.fi/problemset/task/1715

$$\frac{n!}{cnt_a!\, cnt_b!\, cnt_c! \cdots cnt_z!}$$

aab → aab, aba, baa

Problem

You're given an integer **N**.

Find the number of permutations of the first **N** natural numbers such that all the even numbers come together.

[1, 2, ... , N]

N = 6

[1, 2, 3, 4, 5, 6]

[1, $\boxed{2, 4, 6}$, 5, 3]

[1, 4, 2, 6, 3, 5]

$N \longrightarrow [1, 2, \ldots, N]$

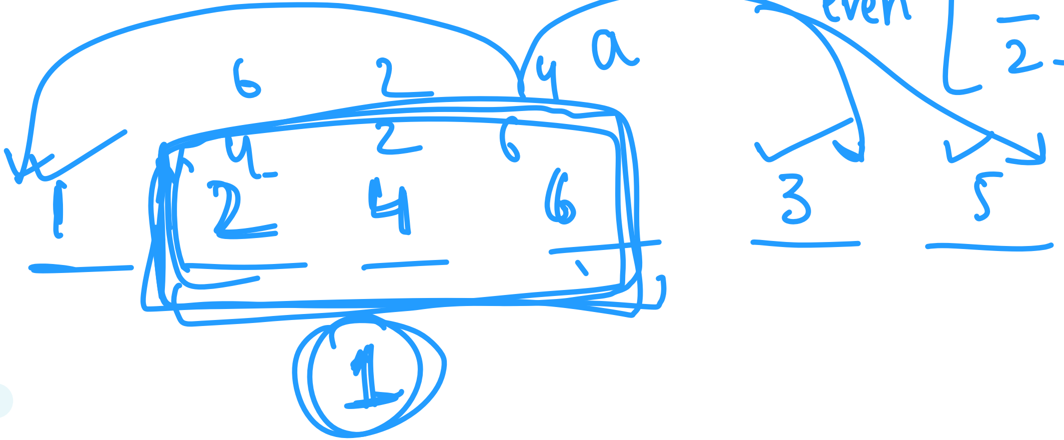
$N \to$ even then $cnt_{even} = N/2$ $cnt_{odd} = N/2$

$ \searrow$ odd then $cnt_{even} = \frac{N-1}{2}$ $cnt_{odd} = \frac{N+1}{2}$

$N=5$

$[\checkmark 1, \checkmark 2, \checkmark 3, \checkmark 4, \checkmark 5]$

$cnt_{even} = \left\lfloor \frac{N}{2} \right\rfloor$ $cnt_{odd} = \left\lceil \frac{N}{2} \right\rceil$



$$\boxed{(cnt_{odd} + 1)! + cnt_{even}!}$$

1 a 3 5

a 1 3 5

_ a _ _

_ _ a _

_ _ _ a

$4!$

odd → 1 3 5 → $3!$

even → 2 4 6 → $3!$

$$cnt_{odd}!$$

$$* \; cnt_{even}!$$

$$* \; (cnt_{odd}+1)$$

$$\Rightarrow (cnt_{odd}+1)! * cnt_{even}!$$

$$(cnt_{even}+1)!(cnt_{odd}!)$$

$cnt_{even}+1$

2, 1, 3, 5, 4, 6

$(cnt_{odd}+1)$